

```

1 #include <stdio.h>
2 #include <unistd.h>
3 #include <string.h>
4 #include <sys/wait.h>
5 #include <time.h>
6
7 int main()
8 {
9     int fd[2];
10    char msg[] = "Hello from Parent";
11    char buffer[100];
12
13    pipe(fd);
14
15    clock_t start = clock();
16
17    int pid = fork();
18
19    if (pid == 0)
20    {
21        close(fd[1]);
22
23        read(fd[0], buffer, sizeof(buffer));
24
25        printf("Child consumed: %s\n", buffer);
26
27        close(fd[0]);
28    }
29    else
30    {
31        close(fd[0]);
32
33        printf("Parent produced: %s\n", msg);
34
35        write(fd[1], msg, strlen(msg) + 1);
36
37        close(fd[1]);
38
39        wait(NULL);
40
41        clock_t end = clock();
42
43        double time = (double)(end - start) / CLOCKS_PER_SEC;
44
45        printf("Communication time: %f seconds\n", time);
46    }
47
48    return 0;
49 }

```

```

yeshwanth@LAPTOP-N3S9K224:~$ gcc q5.c
yeshwanth@LAPTOP-N3S9K224:~$ ./a.out
Parent produced: Hello from Parent
Child consumed: Hello from Parent
Communication time: 0.000758 seconds

```

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/wait.h>
4
5 int main()
6 {
7     int fd[2];
8     pipe(fd);
9
10    int pid1 = fork();
11
12    if (pid1 == 0)
13    {
14        dup2(fd[1], STDOUT_FILENO);
15
16        close(fd[0]);
17        close(fd[1]);
18
19        execlp("ls", "ls", "-l", NULL);
20    }
21
22    int pid2 = fork();
23
24    if (pid2 == 0)
25    {
26        dup2(fd[0], STDIN_FILENO);
27
28        close(fd[1]);
29        close(fd[0]);
30
31        execlp("grep", "grep", ".c", NULL);
32    }
33
34    close(fd[0]);
35    close(fd[1]);
36
37    wait(NULL);
38    wait(NULL);
39
40    return 0;
41 }
```

```

yeshwanth@LAPTOP-N3S9K224:~$ gedit q5b.c

** (gedit:2674): WARNING **: 05:37:20.023: Could not load Peas repository: 1

** (gedit:2674): WARNING **: 05:37:20.023: Could not load PeasGtk repository: 1
yeshwanth@LAPTOP-N3S9K224:~$ gcc q5b.c
yeshwanth@LAPTOP-N3S9K224:~$ ./a.out
-rw-r--r-- 1 yeshwanth yeshwanth    96 Jul 28 04:44 addition.c
-rw-r--r-- 1 yeshwanth yeshwanth   176 Aug 11 05:06 e1.c
-rw-r--r-- 1 yeshwanth yeshwanth   322 Aug 11 05:06 e2.c
-rw-r--r-- 1 yeshwanth yeshwanth   419 Aug 11 05:21 e3.c
-rw----- 1 yeshwanth yeshwanth   167 Aug  3 08:39 e3.c.save
-rw-r--r-- 1 yeshwanth yeshwanth   226 Aug 11 05:21 e4.c
-rw-r--r-- 1 yeshwanth yeshwanth   186 Aug 11 05:21 e5.c
-rw-r--r-- 1 yeshwanth yeshwanth   159 Jul 28 05:05 multi.c
-rw-r--r-- 1 yeshwanth yeshwanth   327 Jul 28 05:08 operations.c
-rw-r--r-- 1 yeshwanth yeshwanth     0 Aug 11 05:00 p1.c
-rw-r--r-- 1 yeshwanth yeshwanth   186 Aug 11 05:00 p4.c
-rw-r--r-- 1 yeshwanth yeshwanth   564 Aug 17 06:54 p5.c
-rw-r--r-- 1 yeshwanth yeshwanth   194 Aug 11 05:03 p6.c
-rw-r--r-- 1 yeshwanth yeshwanth   184 Aug 17 06:53 p7.c
-rw-r--r-- 1 yeshwanth yeshwanth   158 Aug  2 15:02 process_2.c
-rw-r--r-- 1 yeshwanth yeshwanth   242 Aug  2 15:03 process_3.c
-rw-r--r-- 1 yeshwanth yeshwanth   777 Sep  3 05:34 q5.c
-rw-r--r-- 1 yeshwanth yeshwanth   551 Sep  3 05:37 q5b.c
-rw-r--r-- 1 yeshwanth yeshwanth   150 Jul 28 05:01 subtraction.c
-rw-r--r-- 1 yeshwanth yeshwanth  1129 Jul 28 05:23 switchop.c
-rw-r--r-- 1 yeshwanth yeshwanth  1824 Jul 28 05:26 switchope.c
-rw-r--r-- 1 yeshwanth yeshwanth   512 Aug 11 05:24 w3-1.c
-rw-r--r-- 1 yeshwanth yeshwanth    78 Aug 11 05:19 w4-1.c
yeshwanth@LAPTOP-N3S9K224:~$

```